Technical Reproducibility Validation Report

STRait Razor

Version v3
STR / SNP panel

Verification date	2026-11-01
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Result	**Not verified as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/strait-razor-forenseq
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Not verified as documented**
REACHED	Runs + Plausible output, 5/5 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. What STRhub had to do to run this tool

STRhub wrote its own recipe for this tool, an environment and a command of its own rather than the repository's instructions, and ran that. Each item below is something a first-time user following the README would have to work out for themselves, and so a recommendation to the author. This run does not change the tool's label: the label is what happens as it is in the repository.

– Builds with make (falling back to cmake) and copies str8rzr to /usr/local/bin. Instead of: The README's build steps, which name no install location.

– Picks the ForenSeq kit configuration (ForenSeqv1.27.config) and the STRhub fixture for it. Instead of: A configuration and input the README leaves to the user; the repository ships several kit configurations.

With those changes, the run reached: **Runs + Plausible output**.

3. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

4. What the Author Documents as a Known Issue

Quoted from the repository's README at the verified commit. STRhub did not establish any of this by running the tool; it is the author's own note about their own software.

known issues (README line 114)

There's been one computer architecture (windows 7 + Xeon processor) that's caused some issues with the str8rzr.exe executable. In this case, str8rzr would occassionally crash, and I had to recompile it for that computer (and since then it's been fine). Please let me know if you experience problems-- especially crashes-- it'll let me further diagnose the exact problems therein. str8rzr is written in C/C++ with multithreading support using the pthreads library. This new release couples a new search strategy (see algorithm) coupled with a complete redesign of the code-base used to identify short …

5. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · STRait Razor v3 · commit b618e93
$ str8rzr \
-c /opt/strait-razor/ForenSeqv1.27.config \
/data/in/sample.fastq > /data/out/sample.allsequences.txt
```

6. Reproducibility Metadata

Tool	STRait Razor
Version	v3
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-11-01 12:46:33 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/41307968157

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

7. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

8. Output Content Evidence

Output file	sample.allsequences.txt
Format	TSV
Sequence records	816
Distinct STR loci	63
Total reads	3,967
Max depth (single locus)	132

9. Verification Data

This tool ships no demo or test data of its own. STRhub ran the verification using the public reference dataset listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D17S1301 · D18S51 · D19S433 · D1S1656 · D20S482 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D4S2408 · D5S818 · D6S1043 · D7S820 · D8S1179 · D9S1122 · DXS10074 · DXS10103 · DXS10135 · DXS7132 · DXS7423 · DXS8378 · DYF387S1 · DYS19 · DYS385 · DYS389I · DYS389II · DYS390 · DYS391 · DYS392 · DYS437 · DYS438 · DYS439 · DYS448 · DYS460 · DYS461 · DYS481 · DYS505 · DYS522 · DYS533 · DYS549 · DYS570 · DYS576 · DYS612 · DYS635 · DYS643 · FGA · HPRTB · N29insA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · mh16-MC1RB · mh16-MC1RC · vWA

10. Verification Matrix

PASS	Reference dataset	NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

11. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– A container environment and a command written by STRhub, not taken from the repository's instructions; what they do differently is listed under "What STRhub had to do to run this tool".

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

◆ **Runs end-to-end**

STRait Razor v3 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

◆ **Produces plausible output**

The tool generated a structurally valid TSV output file with genotype calls across 63 target forensic STR loci, with a maximum read depth of 132 at DYS392.

◆ **Verdict: Runs**

The tool installed and its run produced its documented output.

◆ **No bundled demo data**

STRait Razor ships no test or demo data of its own. STRhub Verified supplied NIST mds2-2157, Illumina STR (ForenSeq slice, donor NTD01) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

STRait Razor v3 · verified 2026-11-01 · showing the 60 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH		LOCUS	READ DEPTH	
DYS392	170	■■■■■■■■■■■■■■■■■	DYS643	37	■■■
DYS438	117	■■■■■■■■■■■■	HPRTB	36	■■■
D6S1043	111	■■■■■■■■■■■	rs4364205	36	■■■
TH01	106	■■■■■■■■■■	D8S1179	34	■■■
DYS389I	97	■■■■■■■■■■	DXS10135	34	■■■
DYS576	95	■■■■■■■■■	DYF387S1	34	■■■
DYS505	93	■■■■■■■■■	DYS390	34	■■■
D20S482	84	■■■■■■■■	D12S391	31	■■
FGA	81	■■■■■■■■	DYS391	31	■■
D3S1358	69	■■■■■■■	D21S11	30	■■
D2S441	66	■■■■■■■	rs1919550	29	■■
DYS612	63	■■■■■■	D13S317	28	■■
D16S539	61	■■■■■■	rs174570	28	■■
D19S433	59	■■■■■■	rs192655	28	■■
DXS10074	58	■■■■■■	D17S1301	26	■■
D4S2408	52	■■■■■	CSF1PO	24	■■
DXS7132	51	■■■■■	rs1454361	24	■■
D18S51	48	■■■■■	rs8078417	24	■■
rs200354	47	■■■■■	DYS389II	23	■■
DYS385	45	■■■■■	DYS439	23	■■
D22S1045	44	■■■■■	DYS570	23	■■
DXS8378	44	■■■■■	rs1382387	23	■■
DXS7423	43	■■■■■	rs722098	23	■■
rs1109037	43	■■■■■	PentaD	22	■■
DYS437	40	■■■■	rs2814778	22	■■
mh16-MC1RC	40	■■■■	rs876724	22	■■
DYS448	39	■■■■	rs2402130	21	■
D10S1248	38	■■■■	rs4918664	21	■
rs459920	38	■■■■	rs917118	21	■
D7S820	37	■■■■	DYS481	20	■